

Discrete Raman Amplification in the U+X Band (1610-1690nm) Utilizing Twin EDFA ASE Peaks (1535nm and 1565nm) for Forward Pumping with Reduced Noise Generation

Youichi Akasaka⁽¹⁾, Inwoong Kim⁽¹⁾, and Paparao Palacharla⁽¹⁾

⁽¹⁾ 1Finity Americas Inc., 17201 Waterview Parkway, Dallas, TX 75252 youichi.akasaka@fujitsu.com

Abstract U+X band discrete Raman amplification covering 1610nm to 1690nm is demonstrated using twin peaks of C-band EDFA ASE at 1535nm and 1565nm, made by 4m EDF and 13m EDF, respectively. Forward pumping with incoherent ASE achieved a 1dB Lower noise compared to backward pumping.: ©2026 The Author(s)

Introduction

With progress in Hollow Core Fibers, ultra-wide-band transmission—including wavelengths up to 1700 nm—is becoming practical, since it avoids the limitation of infrared absorption loss above 1650 nm in conventional Silica based optical fiber. As a result, the development of optical amplifiers for the U+X band (1610–1690 nm) has become a priority [1].

Thulium and Bismuth doped fiber amplifiers (TDFA, BDFA) can be used for U-band optical amplification but have limitations, such as limited bandwidth and a large gap with the L-band that prevents effective use of fiber bandwidth [2, 3]. NTT proposed using wavelength conversion and C-band/L-band EDFAs to efficiently amplify both U+X band signals with existing technology [4, 5].

On the other hand, amplifiers using nonlinear effects enable flexible wavelength selection and broader bandwidth. Especially, discrete Raman amplifiers have been extensively studied and can be considered as the leading candidate to utilize full U+X band [6, 7].

When employing the ASE of an EDFA as an incoherent pump for U+X band Raman amplification, forward-pumping can reduce noise due to Raman gain medium attenuation. However, the ASE from a C-band EDFA, despite being broadband, concentrates optical power near 1535 nm, effectively creating single-channel pumping with a limited ~30 nm gain band at 3 dB, thus underutilizing the optical fiber bandwidth.

In this study, we proposed a U+X band, low-noise discrete Raman amplifier by using EDFs of two lengths to shift the EDFA's ASE peak by 30 nm, thereby enabling wideband amplification. Forward-pumping with incoherent ASE pumps reduced Raman noise due to fiber loss. Limiting the gain to about 20 dB in the first stage avoided double Rayleigh Scattering (DRS) penalties [8] and then adding a Raman amplifier with WDM-pumped Fiber Bragg Grating (FBG) lasers in the second stage, we have achieved a flat, high-gain and low-noise performance.

Twin Peaks of EDFA ASE for Raman Pumps

Fig.1 shows the EDFA ASE spectra with various EDF lengths (2, 4, 6, 9, 11, and 13m) in (a) log scale and (b) linear scale. Calculation uses Lightera's OASIX9 simulator with HP-19 EDFs. 6m EDF is good for C-band flat gain from log scale, but most of ASE power is generated around 1530nm. 4m EDF has highest power in linear scale and acts as 1530nm single channel pump. When increasing EDF length, ASE spectra shift towards longer wavelength and 13m EDF gives highest ASE peak power around 1560nm, which could act 1560nm single channel pump. By combining these two ASE peaks, which are apart around 30nm, can make wider gain bandwidth in U+X-band with 13THz shift (~110nm longer wavelength). While 6m EDF has both peaks of 1530nm and 1560nm, but in linear scale, they are less than half the power of 4m or 13m. This 3dB lower power has big impact on Raman gain due to nonlinear phenomenon. So we use 4m and 13m EDFs to generate the ASE peaks.

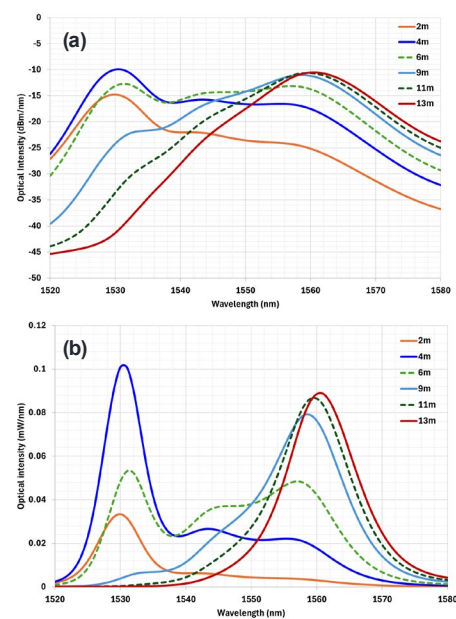


Fig. 1: EDF length dependence on EDFA ASE spectra, (a) log scale, (b) linear scale

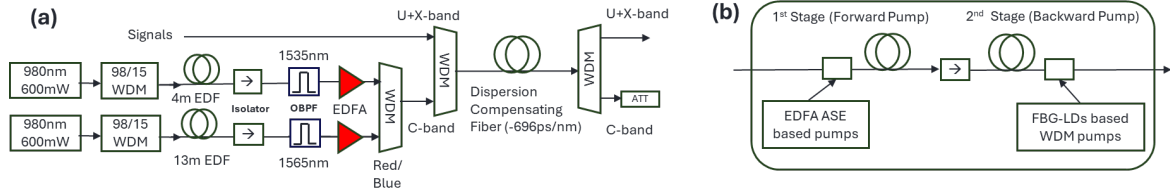


Fig. 2: (a) Experimental Setup for Forward Raman amplifier by twin peaks of EDFA ASE, (b) Configuration of wide band flat gain discrete Raman amplifier for U+X-band

Experimental Setup

Fig. 2(a) shows experimental setup of forward pumping discrete Raman amplifier using twin peaks of EDFA ASE. The twin ASE peaks are individually generated with 980nm pump (Lumentum) of 600mW each and coupled into 4m EDF and 13m EDF for 1530nm and 1560nm peaks, respectively. We used Lightera's HP-19 EDF for both EDFAs. After isolator and optical band pass filter (OBPF) of 1535nm and 1565nm, the ASE peaks are amplified by EDFA individually to control flatness of the gain profile. ASE peaks are 5nm shifted longer by the OBPFs for covering more of the X-band. Output power of each peak is 28.4dBm (692mW) and 28.8dBm (759mW), respectively. After Red/Blue WDM filter, the combined twin peaks, shown in Fig. 3, are injected into Raman gain medium of dispersion compensating fiber (DCF) with signal sources, such as, a noise source (EXFO FLS5800 (-30dBm/0.1nm RB)), or a tuneable laser source (Agilent 81640A (-30dBm/ch)). The DCF has -696ps/nm/km of chromatic dispersion with 7.7km length and 7.1dB attenuation. As Signal / Pump WDM has slightly higher attenuation in U+X-band, evaluation of net gain and net NF used input and output signal powers before and after DCF. Fig. 2(b) shows configuration of wide band flat gain discrete Raman amplifier for U+X-band. The first stage uses the twin EDFA ASE peaks based forward pumped Raman amplifier, and the second stage uses FBG-LDs based WDM pumps based backward pumped Raman amplifier. The second stage net gain is simulated with the configuration of 5km DCF and 4WDM pumps (500mW at 1500nm, 500mW at 1510nm, 70mW at 1530nm, and 60mW at 1585nm).

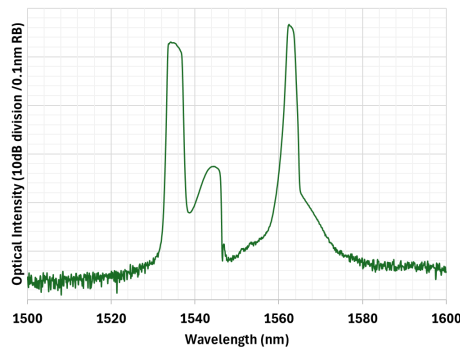


Fig. 3: Twin EDFA ASE peaks incoherent pump

Raman Gain Profile

Fig. 4 shows the Raman gain profile with noise source as signal. Net gain (Solid blue line) is calculated from measured output spectra – input spectra to DCF – noise power without signal noise source. More than 10dB net gain is achieved with bandwidth is >80nm from 1610nm to 1690nm. Net gain of individual EDFA ASE peaks is also measured and shown in dotted sky-blue line (1535nm) and solid orange line (1565nm). Sum of those 2 curves is also shown in dotted red line. We observe that the net gain from the twin peaks has flatter and wider gain profile than that of sum of 2 gain spectra, especially at wavelength >1680nm. This indicates that the energy of the 1535nm peak is transferred to 1565nm peak inside of the Raman gain medium. 45nm (1640nm-1685nm) gain bandwidth of >18dB is achieved, if this amplifier is supposed to be use as single stage discrete Raman amplifier.

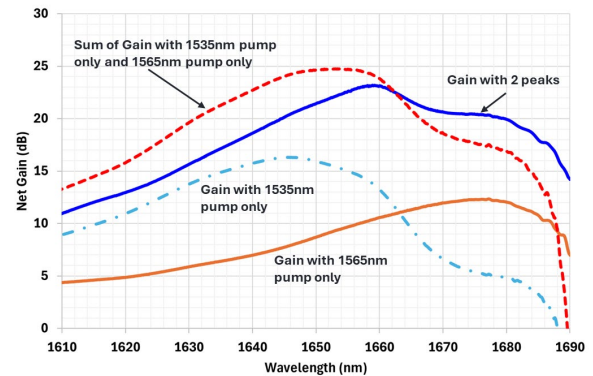


Fig. 4: Raman gain profile by Twin EDFA ASE peaks (Solid Blue line), 1535nm peak only (dotted sky-blue line), 1565nm peak only (Solid Orange line), and Sum of 1535nm pump only & 1565nm pump only (dotted red line)

Noise Performance Evaluation of the First Stage

To evaluate first stage amplifier's noise performance, net gain and noise figure (NF) of single channels from 1636nm to 1641nm (1nm increment) were measured with forward pumping or backward pumping using the same twin EDFA ASE peaks as pump source.

Fig. 5 shows the net gain and NF of forward pumping and backward pumping configuration of the first stage Raman amplifier with the same

pump unit. Due to the availability of tuneable laser source, measured wavelengths are not in the maximum gain region. Forward pumping provides 1.1dB to 2.4dB higher net gain than backward pumping, due to pump and fiber attenuation interaction. NFs for forward pumping are 1dB to 1.34dB lower than for backward pumping because the gain and noise generation occurs near the fiber input, causing greater attenuation of noise along the fiber.

Thus, forward pumping is advantageous and suitable for discrete Raman amplification, as shown by these measurements.

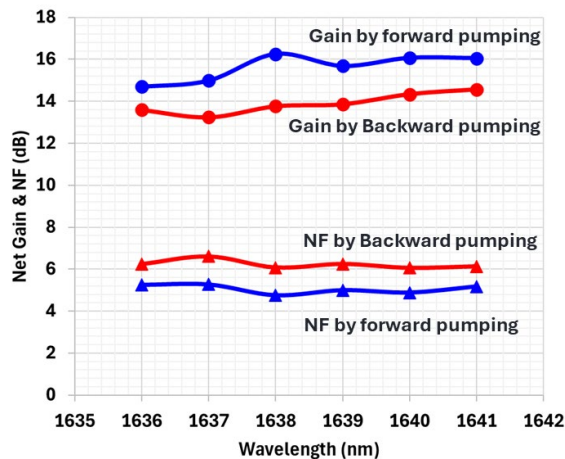


Fig. 5: Net gain and NF of single channel sweep of the first stage.

Wideband Flat Gain Low Noise Discrete Raman Amplifier for U+X-band

As shown in Fig. 2(b), the two stages configuration comprises a first stage with forward pumping using twin EDFA ASE peaks pump source, followed by a second stage employing backward pumping with FBG stabilized semiconductor LDs for WDM pumping. This arrangement is suitable for achieving wideband, flat gain, and low noise performance in discrete Raman amplifiers.

Dividing amplification into two stages keeps the required net gain for each stage below ~20dB, effectively reducing double Rayleigh scattering penalties. Using forward incoherent pumping generates less noise than backward pumping, and twin EDFA ASE peaks expand the gain bandwidth to 80nm. WDM pumping in the second stage flattens the gain across the bandwidth by adjusting pump powers for each wavelength. Fig. 6 shows the total net gain of approximately 30dB (+/- 3dB) across the 80nm bandwidth from 1610nm to 1690nm in the proposed 2-stage system. The estimated noise figure (NF) of this discrete Raman amplifier is about 5dB, as the first stage provides over 10dB net gain, and the second stage contributes only a maximum 10% effect on the overall NF.

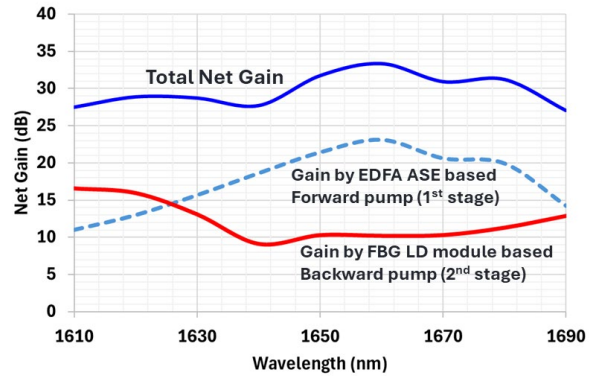


Fig. 6: Net Gain of 1st stage, 2nd stage, and total net gain.

Conclusion

We have demonstrated discrete Raman amplification in the U+X band, covering wavelengths from 1610nm to 1690nm, using twin peaks of ASE at 1535nm and 1565nm from C-band EDFAs, created with 4m and 13m EDFs, respectively. Forward pumping achieved a 1dB lower noise figure due to the incoherency of the ASE compared to backward pumping. By combining this with an additional stage of backward pumping that utilizes FBG-stabilized semiconductor LDs as a WDM pump source, the result is a wide and flat net gain—about 30dB—from 1610nm to 1690nm, and a low noise figure (~5dB). This discrete Raman amplifier can be deployed in future multi-band transmission applications.

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References

- [1] M. Petrovich, E. N. Fokoua, Y. Chen, H. Sakr, A. I. Adamu, R. Hassan, D. Wu, R. F. Ando, A. Papadimopoulos, S. R. Sandoghchi, G. Jasion, & F. Poletti, "Broadband optical fiber with an attenuation lower than 0.1decibel per kilometre," *Nature Photonics*, vol. 19, pp. 1203-1208, 2025, DOI:10.1038/s41566-025-01747-5
- [2] S. Chen, Y. Jung, S. U. Alam, R. Sidharthan, D. Ho, S. Yoo, D. J. Richardson, & J. M. O. Daniel "Ultra-short Wavelength Operation of Thulium-doped Fibre Amplifier in the 1628-1655nm Waveband", *ECOC2018*, 2018, DOI: 10.1109/ECOC.2018.8535134
- [3] S. V. Firstov, K. E. Riumkin, S. V. Alyshev, V. F. Khopin, A. M. Khagai, & M. A. Melkumov, "Temperature-dependent characteristics of bismuth-doped fiber amplifier operating in a 1720-nm band", in *2017 Opto-Electronics and Communications Conference (OECC) and Photonics Global Conference (PGC)*, Singapore, 2017, DOI: 10.1109/OECC.2017.8114878
- [4] C. Guo, A. Shamshooli, Y. Akasaka, T. Ikeuchi, & M. Vasilyev, "Noise Figure Study for a 3-Stage Hybrid Amplifier Using Parametric Wavelength Converters and EDFA", *IEEE Photonics Technology Letters*, Vol.33, Issue 16, pp872-875, 2021. DOI: 10.1109/LPT.2021.3083420

- [5] S. Shimizu, K. Kimura, A. Kawai, M. Abe, S. Konno, T. Kazama, M. Nakamura, F. Hamaoka, K. Enbutsu, T. Kashiwazaki, M. Nagatani, H. Wakita, Y. Shiratori, H. Yamazaki, H. Takahashi, T. Umeki, T. Kobayashi, Y. Miyamoto, "ISRS-Supported Ultra-Wideband Transmission in S, C, L, U, and Extremely Long Wavelength Bands with PPLN-Based OPA/EDFA Hybrid Repeater", *Journal of Lightwave Technology*, Vol. 44, No. 3, pp1139-1150, 2026.
<https://doi.org/10.1109/JLT.2025.3609452>
- [6] T. Kato, H. Irie, H. Muranaka, Y. Tanaka, Y. Akiyama, & T. Hoshida, 'U-band Transmission of Real-Time 200-Gb/s Signal Co-Propagating with C+L-band WDM Signal', *OFC2023*, Th3F.6, 2023.
[10.1364/OFC.2023.Th3F.6](https://doi.org/10.1364/OFC.2023.Th3F.6)
- [7] B. J. Puttnam, R. S. Luis, I. Phillips, M. Tan, A. Donodin, & D. Pratiwi, "High Data-Rate OESCLU-Band Transmission," *Journal of Lightwave Technology*, Vol. 43, Issue 23, 2025. DOI: [10.1109/JLT.2025.3543448](https://doi.org/10.1109/JLT.2025.3543448)
- [8] P. Parolari, L. Marazzi, L. Bernardini, & M. Martinelli, "Double Rayleigh Scattering Noise in Lumped and Distributed Raman Amplifiers," *Journal of Lightwave Technology*, vol. 21, no. 10, pp. 2224-2228, 2003.
DOI: [10.1109/JLT.2003.817705](https://doi.org/10.1109/JLT.2003.817705)